Project Proposal: Crowd-Driven Visual Generation with Conditional Diffusion Models

Template version V0, 03/25/2026 · **Course** Generative AI – Principles and Applications, CCE IISc

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1. Problem Statement

Problem. How can a single, coherent image be generated by conditioning a diffusion model on a *distribution of thousands of heterogeneous inputs* — words, emojis, and doodles contributed by a live audience? We frame this as a *many-to-one* conditioning problem.

Relevance. *Real-world:* audience participation at live events is passive (flashlights, applause); no tool lets a crowd co-create the on-screen visuals — during a song, fans might submit one word, an emoji, or a quick doodle. *Research gap:* conditional generation is almost universally one-input-to-one-output — a class label, a text prompt, or a single image. Conditioning *one* generation on an unordered set of *thousands* of heterogeneous inputs, coherently, is under-explored, and it is not achievable by a single prompt without an explicit, learned aggregation step (a general model accepts one prompt, not a crowd). The scope for this course project is the model and a simulated-crowd study, not a live stadium deployment.

Domain. Deep generative models — conditional latent diffusion (DDPM), with VAE/VQ-VAE representation learning, permutation-invariant set aggregation and event-metadata conditioning, a GAN baseline, and sampling (DDPM/DDIM). Course modules: Diffusion Models and Generative Models.

2. Proposed Approach

Core idea. A conditional latent-diffusion pipeline that maps a *crowd* of inputs to *one* artwork: heterogeneous inputs are embedded into a shared latent space, an aggregation module collapses the set into a *unified conditioning representation* — fused with structured event metadata (artist, song, genre, venue) — and a conditional DDPM with classifier-free guidance synthesises the artwork. The central contribution is a systematic study of *permutation-invariant aggregation strategies* for diffusion conditioning, benchmarked against a GAN.

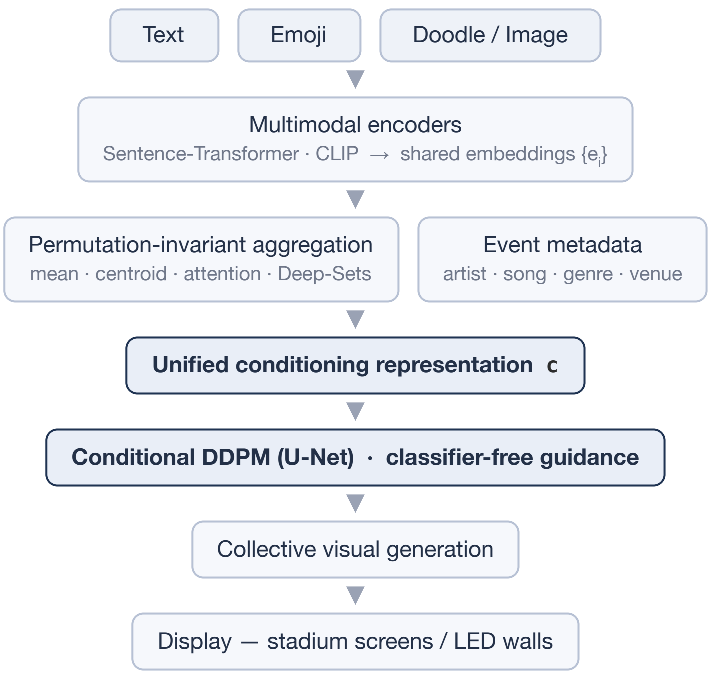

Figure 1. The crowd-driven generation pipeline: many heterogeneous audience inputs collapse into a single unified conditioning representation (fused with event metadata) that guides a conditional diffusion model to synthesise one coherent visual.

Technical plan.

- *Encoding*. Each input $i_i \to e_i \in \mathbb{R}^d$ in a shared space — words and emojis via a text encoder, doodles via a CNN/VQ-VAE image encoder.
- *Aggregation (core)*. Map the unordered set $\{e_1 \dots e_n\} \to c$, a *unified conditioning representation*. Since crowd-input order is meaningless, the module must be permutation-invariant. Compared: (i) mean-pool $c = (1/N)\Sigma e_i$ (naïve baseline); (ii) cluster-centroid mixture (k-means, capturing multi-modality); (iii) attention pooling (Set-Transformer PMA); (iv) Deep-Sets $c = \rho(\Sigma\varphi(e_i))$.
- *Event-aware conditioning*. Fuse the aggregated representation with structured event metadata (artist, song, genre, venue, theme) — via concatenation or cross-attention, compared as an ablation — so the output reflects both audience sentiment and event context.
- *Diffusion*. Latent-space DDPM with the noise-prediction objective $L = \mathbb{E}_{k,x_0,\varepsilon} \|\varepsilon - \varepsilon\theta(x_t, t, c)\|^2$ and forward process $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{a}_t}\ x_0, (1-\bar{a}_t)I)$; the condition c enters the U-Net via cross-attention / FiLM.
- *Training signal*. The conditional model is trained on single image-level embeddings (each training image paired with its own CLIP/caption embedding); crowd aggregation is applied only at inference, composing many embeddings into one — so no crowd–artwork paired data is required.
- *Classifier-free guidance*. Train with condition-dropout ($c \to \varnothing$); sample $\hat{\varepsilon} = \varepsilon\theta(x_t,\varnothing) + w\cdot(\varepsilon\theta(x_t,c) - \varepsilon\theta(x_t,\varnothing))$, where w dials the strength of the crowd theme.
- *Sampling*. DDPM (ancestral) versus DDIM (few-step) for near-real-time, per-song generation.
- *Animation*. Spherical interpolation (slerp) between successive songs' conditioning vectors c yields a smoothly evolving artwork.

Project type. Working demo and benchmark study (aggregation-strategy comparison).

Models. VAE/VQ-VAE (from scratch); conditional DDPM (U-Net) with classifier-free guidance; DDIM sampler; permutation-invariant aggregators (mean, k-means, Set-Transformer PMA, Deep-Sets); conditional GAN (baseline); CLIP for theme-fidelity scoring.

Datasets / benchmarks. (a) *Visual domain:* a public art dataset — WikiArt (abstract subset) or a curated abstract/poster set — for training the VAE and diffusion model; FID computed against a held-out split. (b) *Crowd inputs:* a crowd-response simulator producing theme-related words, emojis, and doodles with controllable diversity, using QuickDraw (doodles) and public word/emoji embeddings.

What is novel. Standard conditional generation is one-input-to-one-output; we address the unexplored *many-to-one* setting — conditioning a single coherent diffusion generation on thousands of heterogeneous inputs — and provide a systematic comparison of permutation-invariant aggregation strategies for diffusion conditioning in the crowd/set-conditioning setting.

3. Related Work

#	REFERENCE (TITLE, AUTHORS, YEAR)	HOW OUR PROJECT DIFFERS
1	*Denoising Diffusion Probabilistic Models*, Ho, Jain & Abbeel, 2020	Same DDPM backbone, but conditioned on an aggregated crowd set rather than a single label or image.
2	*Classifier-Free Diffusion Guidance*, Ho & Salimans, 2022	We apply guidance to a set-aggregated conditioning vector and study the guidance dial for crowd-theme strength.
3	*High-Resolution Image Synthesis with Latent Diffusion Models*, Rombach et al., 2022	We use latent diffusion for efficiency, but condition on a learned aggregation of thousands of user inputs, not text or class.
4	*Deep Sets*, Zaheer et al., 2017	We repurpose permutation-invariant set pooling as the conditioning mechanism for generative diffusion.

#	REFERENCE (TITLE, AUTHORS, YEAR)	HOW OUR PROJECT DIFFERS
5	*Set Transformer*, Lee et al., 2019	We adapt attention-based set pooling (PMA) to summarise a crowd into a diffusion condition, benchmarked against simpler aggregators.
6	*Compositional Visual Generation with Composable Diffusion Models*, Liu et al., 2022	They compose a few concepts via score algebra; we aggregate thousands of heterogeneous inputs into one permutation-invariant condition and systematically compare aggregation strategies.

Baseline: Conditional Generative Adversarial Nets, Mirza & Osindero, 2014 — used for the diffusion-versus-GAN comparison in §5.

4. Expected Deliverables

- **Code repository.** Modular and reproducible — crowd simulator, VAE/VQ-VAE, conditional DDPM, aggregation module (all four strategies), GAN baseline, and evaluation scripts, with config files, fixed seeds, and a README giving exact run instructions.
- **Report.** Methodology (with derivations), the aggregation-strategy study, the diffusion-versus-GAN comparison, interpolation/animation, ablations, and honest limitations.
- **Demo.** A recorded video of a simulated crowd producing an evolving artwork across several songs. A live or deployed interface is a stretch goal.

5. Evaluation Plan

Each research question maps to a metric and a baseline.

RQ	QUESTION	BASELINE(S)	METRIC(S)
RQ1	Which aggregation yields coherent art?	mean-pool (naïve floor)	coherence (human-rated single-scene quality + a no-reference quality proxy), theme fidelity (CLIP input↔output), FID, diversity
RQ2	Diffusion versus GAN for many-to-one?	conditional GAN	FID, coherence, CLIP fidelity
RQ3	Do interpolated conditions give smooth animation?	linear versus slerp	perceptual path length (PPL), qualitative reel
RQ4	DDPM versus DDIM speed/quality?	DDPM (full steps)	FID versus number of steps, latency (inference time per artwork)

Additional baseline. An LLM-summary → text-to-image pipeline (aggregate the crowd into a text prompt, then generate) tests whether a learned aggregation improves over prompt-summarisation.

Ablation studies. Aggregation strategy (mean, centroid, Set-Transformer, Deep-Sets); number of inputs `N`; guidance scale `w`; clustering `k`; and the conditioning mechanism (cross-attention versus FiLM). Every claim is tied to these metrics — for example, "attention pooling improves coherence" is supported by the coherence and FID deltas relative to the mean-pool baseline.

Qualitative evaluation. A small user study assesses visual coherence, perceived emotional relevance, how well the visual represents audience participation, and overall appeal — complementing the quantitative metrics, since FID alone does not indicate whether a visual reflects the crowd's collective input.

6. Timeline

WEEKS	PLAN	MILESTONE / OUTPUT
1–2	*Setup & literature review* — repository and environment setup; review DDPM, classifier-free guidance, latent diffusion, Deep-Sets and Set-Transformer, and conditional GANs; finalise datasets and the crowd-simulator specification.	project scaffold; reviewed related work
3–5	*Data & representation* — build the crowd-response simulator (words, emojis, doodles; controllable size `N` and diversity); prepare the visual dataset; implement input encoders into a shared space; train the base VAE/VQ-VAE.	working latent space and simulator; reconstructions; latent visualisation
6–8	*Conditional diffusion core* — implement the conditional DDPM (U-Net) in latent space (noise-prediction loss, timestep embedding); add single-input conditioning with classifier-free guidance; implement the DDIM sampler.	working conditional generator; FID and control-fidelity (single-input)
9–11	*Aggregation study (core contribution)* — implement all four aggregation strategies; run the many-to-one experiments; measure coherence, CLIP fidelity, FID, and diversity.	aggregation-strategy results (RQ1)
12–13	*Baseline & sampling* — implement the conditional GAN baseline and run the diffusion-versus-GAN comparison; benchmark DDPM versus DDIM (quality versus number of steps, latency).	RQ2 and RQ4 results
14–15	*Animation & ablations* — interpolation animation via slerp with perceptual path length; ablations (`N`, `w`, `k`, conditioning mechanism); recorded demo; reproducibility polish.	RQ3 results; demo; reproducible repository
16	*Final report* — writing, figures, analysis, and limitations.	complete report